MECHANICS OF SOLIDS PROJECT

ROTATING-SQUARES AUXETICS

FROM RIGID TILES TO A SIMULATED TPU LATTICE

MOS PROJECT 2026–27 · ROTATING-SQUARES STRAND

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Summary

Square tiles joined at their corners counter-rotate under load and open or close the gaps between them, so the lattice grows or shrinks equally in every direction: an ideal Poisson's ratio of exactly −1. The strand develops this mechanism in two stages.

A first, stiff PLA design (a = 10 mm, printed at $\theta_0 = 15°$) is analysed with three models — rigid squares on pins, compliant hinges by an energy method, and a beam finite-element model written for the project — which agree to 10^{-5} ($\nu = -0.9915$, $E_{eff} = 23.7$ MPa). A failure analysis shows the ligaments yield at under 1 % lattice strain. A softer TPU specimen (a = 8 mm, $\theta_0 = 30°$) is then designed and simulated in ANSYS: under 5 mm of compression it carries 29.9 N and contracts sideways with an effective Poisson's ratio of **−0.698** — clearly auxetic, but well short of the ideal.

PART ONE

1

THE MECHANISM AND ITS MODELS

Rigid squares, compliant hinges, and an independent check

1.1 Developments in rotating-squares auxetics

After Lakes made the first deliberately auxetic foam and Evans and co-workers named the class, the search turned to mechanisms simple enough to analyse exactly. Grima and Evans proposed the most compact one: rigid squares connected at their corners, which counter-rotate under load [1]. Because every square turns by the same angle, the expansion is the same in every direction and the Poisson's ratio is exactly −1. The idea was generalised to semi-rigid units [2] and rectangles [3], and later to rotating rigid units of many shapes [4]. Reviews place it among the canonical auxetic mechanisms [5]; hierarchical versions [7] and stent-like tubes [6, 9] show where it can be used, and simulations on metal lattices [8] show how real units depart from the ideal.

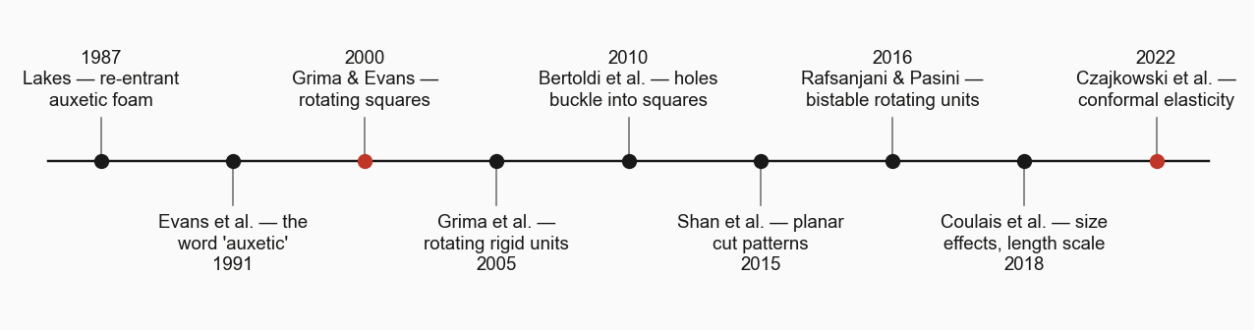


Fig. 1 A short timeline of rotating-unit auxetics.

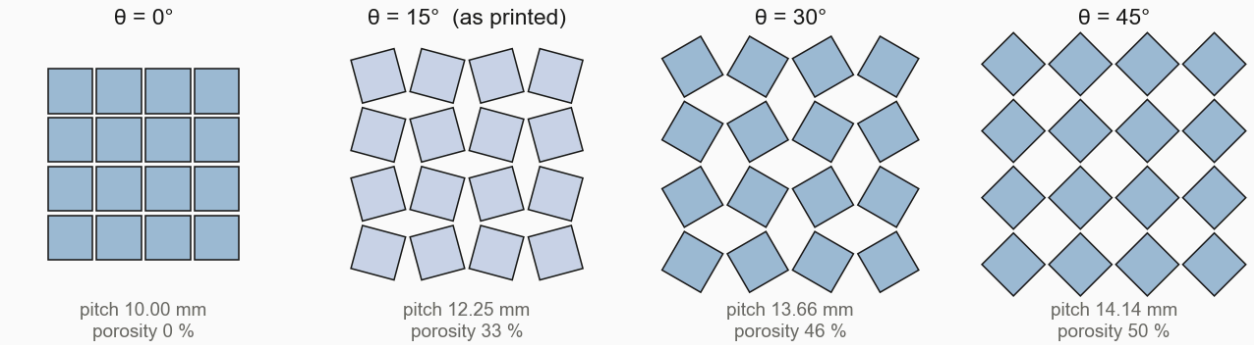


Fig. 2 The ideal lattice at four rotation angles, at the same scale (PLA design convention: θ = 0 closed, θ = 45° fully open). The pitch grows from a to a√2 and the open area from 0 to 50 %.

1.2 Rigid squares on ideal pins

Squares of side a form a checkerboard: one type rotates by −θ, the other by +θ, and every corner is pinned to one neighbour. Requiring that the shared hinge is reached from both square centres (Fig. 3) closes a vector loop and gives the pitch — the same in x and in y:

$$p(\theta) = a\,(\cos\theta + \sin\theta) \tag{1}$$

Because the pitch is one function of one angle in both directions, every rotation increment stretches both equally:

$$\nu_{xy} = -\frac{d\varepsilon_y}{d\varepsilon_x} = -1, \qquad \frac{dp}{d\theta} = a(\cos\theta - \sin\theta) = 0 \text{ at } \theta = 45^\circ \tag{2}$$

The result holds for any size, material and strain. The second relation marks the *open lock*: at 45° the pitch is at its maximum and the lattice cannot stretch further; at θ = 0 the squares meet edge to edge and it cannot shrink. The model is a one-degree-of-freedom mechanism, so it predicts shape but not stiffness. The solid area fraction is $\sin 2\theta/(1+\sin 2\theta)$ open, reaching 50 % porosity at 45°.

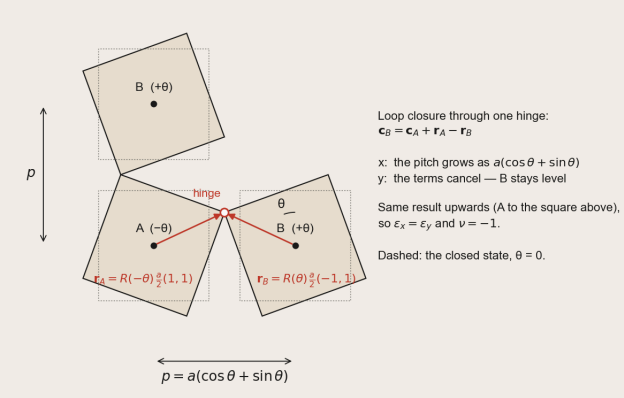


Fig. 3 Vector-loop closure through one hinge. The vertical components cancel, so neighbouring squares stay level and the pitch is identical in x and y.

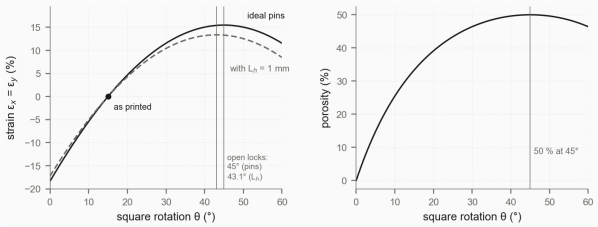


Fig. 4 Strain and porosity against rotation for the PLA design printed at θ_0 = 15°; the open lock lies at +15.5 %.

1.3 Compliant hinges: the energy method

A printed hinge is a short ligament clamped into both squares, of length L_h, width w and depth t. It acts as two springs: bending $k_r = EI/L_h$ for relative rotation and stretching $k_a = EA/L_h$, with $I = tw^3/12$. Linearising the loop of Fig. 3, the stretch of a ligament is

$$\delta = p_0\,\varepsilon - b\,\Delta\theta, \qquad b = a(\cos\theta_0 - \sin\theta_0) \tag{3}$$

where the lever arm b converts square rotation into hinge opening. Minimising the energy of a repeating cell over the internal rotation — rotation and stretching act as springs in series — gives

$$\nu = -\frac{1}{1+8\rho}, \qquad E_{\text{eff}} = \frac{8\,k_r}{t\,b^2\,(1+8\rho)}, \qquad \rho = \frac{w^2}{12\,b^2} \tag{4}$$

Thin ligaments (ρ → 0) recover v = −1; thick ones let stretching take part and weaken the effect. For the PLA design (a = 10 mm, L_h = 1 mm, w = 0.8 mm, t = 5 mm, θ_0 = 15°) this gives **v = −0.9915** and **E_{eff} = 23.7 MPa**, about 148 times softer than PLA. With the ligament arms rotating with the squares, the open lock moves from 45° to 43.1° (+13.4 % strain).

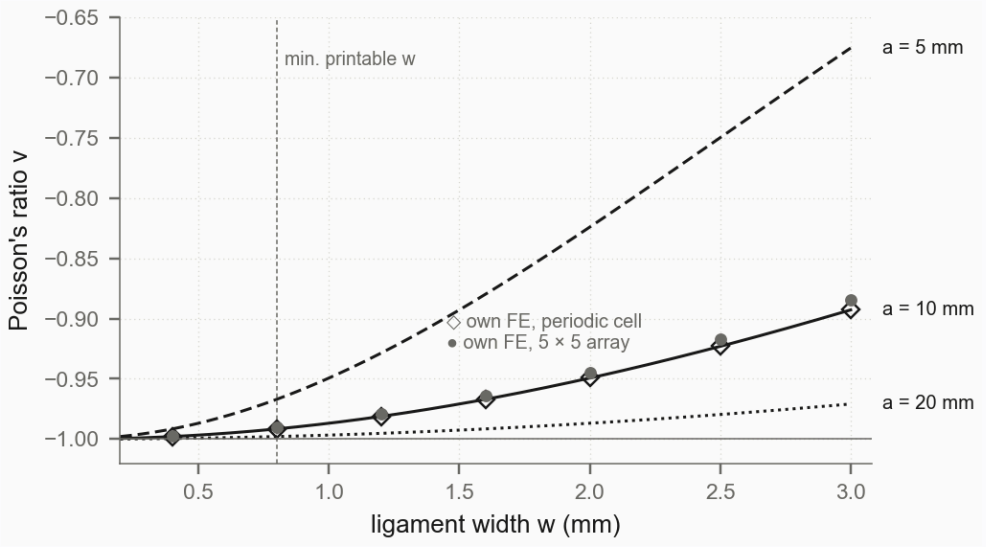


Fig. 5 Poisson's ratio against ligament width for three square sizes. Wider ligaments stretch more and pull v away from −1.

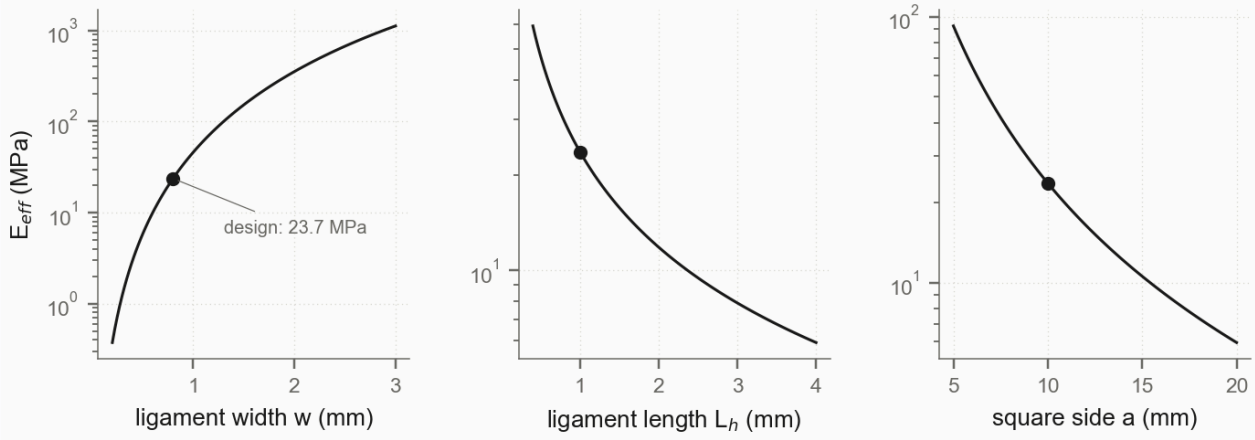


Fig. 6 One parameter at a time around the design point: stiffness rises steeply with ligament width and falls with ligament length and square size, while v barely moves.

1.4 An independent beam finite-element check

To test the energy method without reusing any of its assumptions, a 2D frame finite-element solver was written for the project. Each square is a rigid star of beams; each ligament is four Euler–Bernoulli beam elements. Two versions were run: a periodic cell (the infinite lattice) and finite arrays of 5 × 5 and 13 × 13 squares, in tension and compression.

Model	ν	E_{eff} (MPa)	Δν vs energy method
Rigid squares + pins	−1	—	—
Energy method (infinite)	−0.99154	23.691	reference
Beam FE, periodic cell	−0.99153	23.691	0.001 %
Beam FE, 5 × 5 array	−0.99036	23.682	0.12 %
Beam FE, 13 × 13 array	−0.99003	23.684	0.15 %

Table 1. Three independent descriptions of the PLA design.

The periodic FE cell reproduces the energy method to about 10^{-5} in both ν and stiffness — a strong check on the algebra of Eq. (4). The finite arrays differ by only about 0.1 %: for this stiff design, edges barely matter. Across ligament widths from 0.4 to 3 mm, every model follows the same curve (Fig. 8).

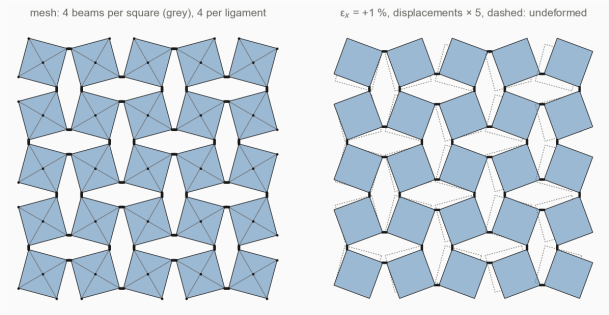


Fig. 7 The beam mesh (four elements per ligament) and the deformed 5 × 5 array at +1 % strain, exaggerated.

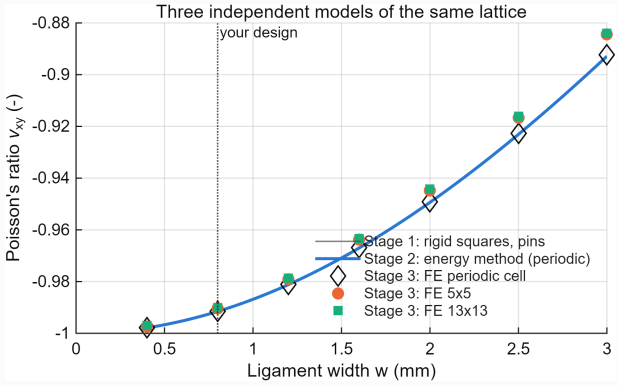


Fig. 8 Poisson's ratio from all models against ligament width; the design point is w = 0.8 mm.

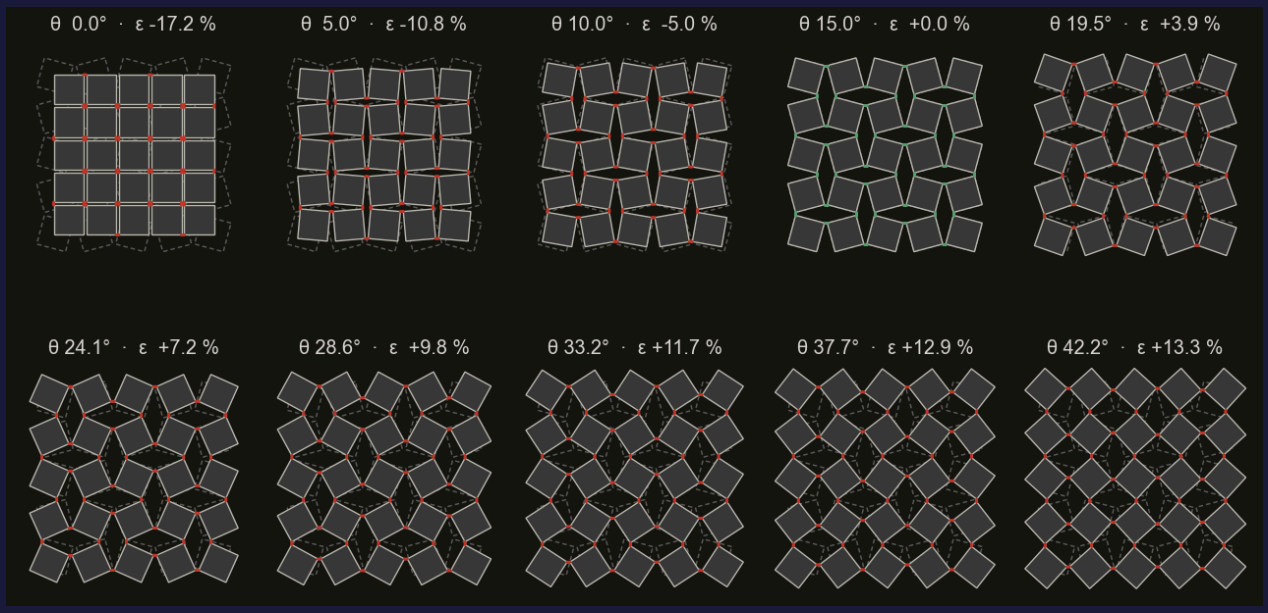


Fig. 9 The PLA lattice driven from closed to fully open, with the lattice strain at each frame.

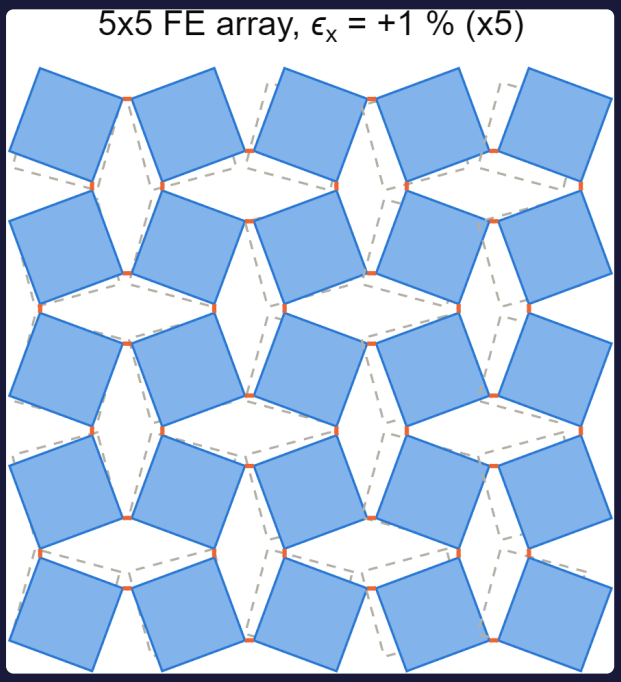


Fig. 10a Beam FE, 5 × 5 array at +1 %.

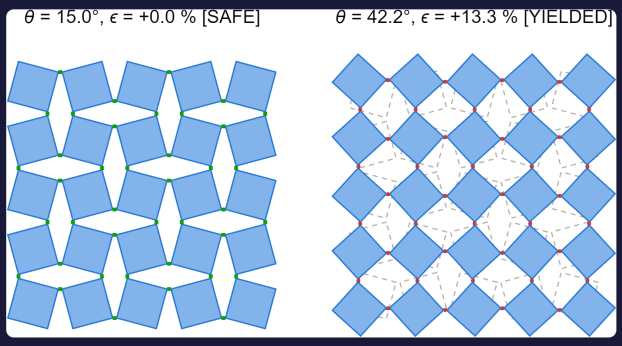


Fig. 10b As printed (θ_0 = 15°, safe) and near the open lock (42.2°, yielded).

1.5 Where the lattice fails

The peak ligament stress combines bending at constant moment with the axial force needed for equilibrium:

$$\sigma_{\max} = K_t \left(\frac{E\, w\, \Delta\theta}{L_h} + \frac{\sigma_x\, p_0}{w} \right) \tag{5}$$

About 93 % of it is bending. Five limits were compared on one axis (Fig. 12): first yield, the open lock, square-to-square contact, and in-plane buckling from a linear eigenvalue analysis of the FE model.

Limit	Lattice strain
First yield (tension / compression)	+0.87 % / −0.90 %
Open lock	+13.4 %
Square contact	−21.0 %
In-plane buckling (linear eigenvalue)	−272 %
Safe range (SF 1.5, edge factor 1.065)	**−0.56 % to +0.55 %**

Table 2. Failure limits of the PLA design.

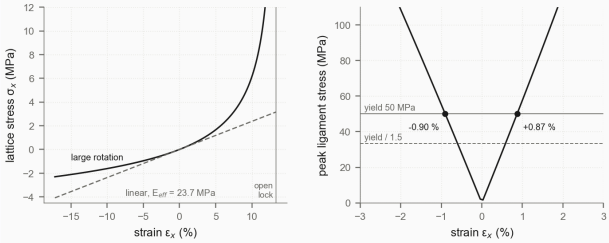


Fig. 11 Lattice stress and peak ligament stress against strain; the ligaments reach PLA yield at about ±0.9 %.

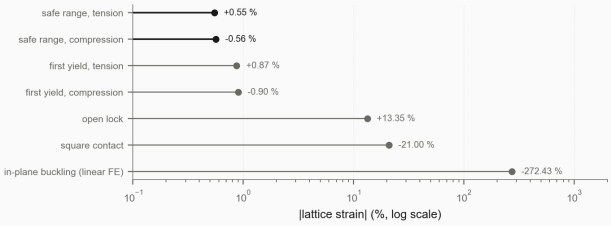


Fig. 12 All limits on one log axis. Yield governs by more than an order of magnitude.

Yield governs. The kinematic locks and buckling lie far beyond the strain at which the PLA ligaments first yield, so large auxetic motion in PLA needs plastic hinges. Longer ligaments ($L_h \approx 3$ mm) roughly triple the elastic range — and a softer, more ductile material removes the problem altogether, which motivates Part 2.

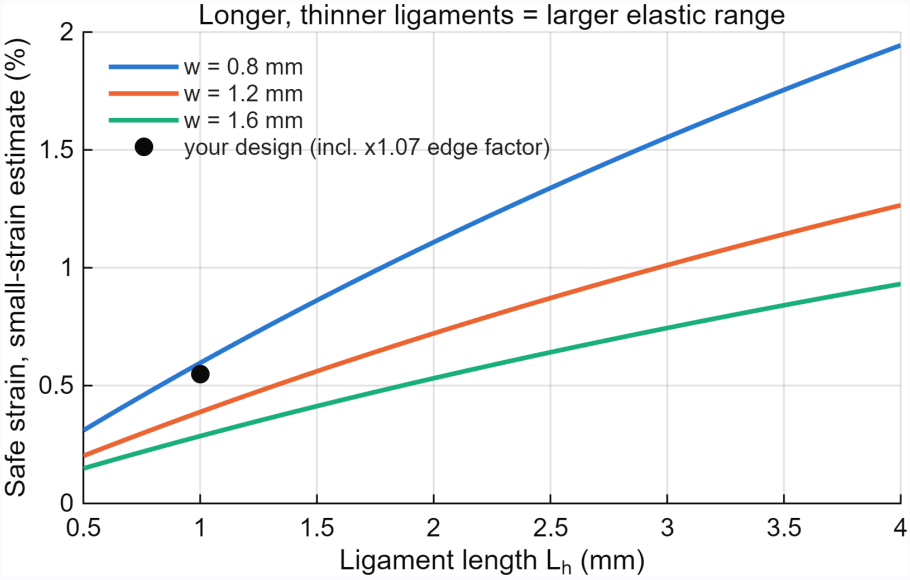


Fig. 13 Safe small-strain range against ligament length for three ligament widths. Longer, thinner ligaments enlarge the elastic window.

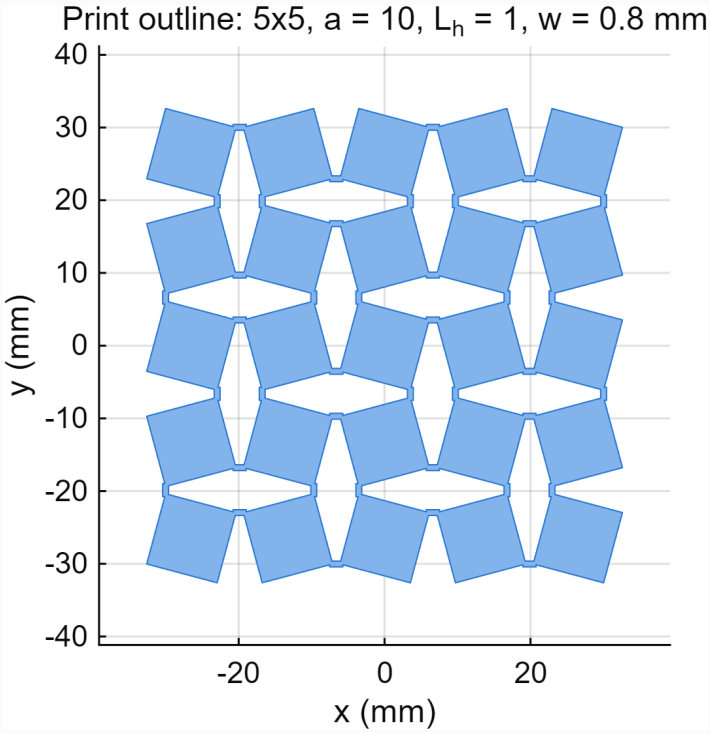


Fig. 14 Print outline of the 5 × 5 PLA design (a = 10, L_h = 1, w = 0.8 mm).

PART TWO

2

THE TPU SPECIMEN IN ANSYS

A softer lattice, a geometry audit, and a 5 mm compression test in simulation

2.1 A softer redesign

The PLA study showed that a stiff material confines the mechanism to under 1 % strain. The second design is a TPU specimen meant to move through the mechanism elastically: 36 squares (3 × 6 of each orientation), a = 8 mm, corner chamfer d = 1.2 mm, ligament width h = 1 mm, depth t = 50 mm, printed at **θ_0 = 30°**, with end blocks giving a 50 × 50 × 100 mm envelope and a 44.37 × 85.94 mm active region. Its angle is measured the other way: here **θ = 45° is closed and θ = 0° fully open**, so

$$\varepsilon_x = \varepsilon_y = \frac{\cos\theta}{\cos\theta_0} - 1, \qquad \phi_s = \frac{1}{2\cos^2\theta}, \qquad E_{\text{eff}} = \frac{E\,h^3}{3\,L_h\,a^2\sin^2\theta_0} = 0.369\ \text{MPa} \tag{6}$$

with $L_h = \sqrt{2}\,d\cos\theta_0 = 1.47$ mm and a provisional E = 26 MPa. The ideal limits from 30° are +15.5 % (open) and −18.4 % (closed). Note $L_h/h \approx 1.47$: these ligaments are not slender, so the hinge model is approximate.

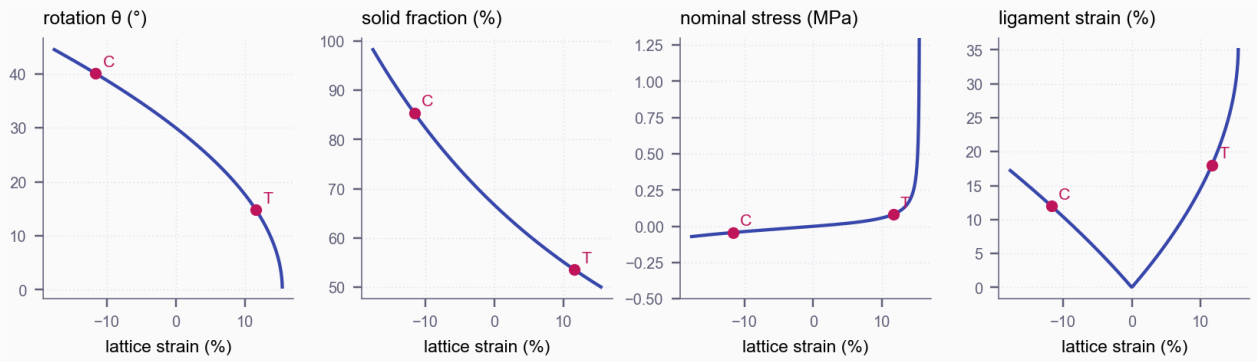


Fig. 15 Analytical relations for the TPU specimen, recomputed from the hinge model. C and T mark 10 mm of active-region compression and tension: 40.1° / 85.4 % solid / 94.7 N, and 14.8° / 53.5 % solid / 182.8 N, with ligament strains of 12.0 % and 18.0 %.

2.2 Geometry audit

Before simulating, the supplied STEP file was imported with Open CASCADE and measured directly. It declares millimetres and contains one fused solid with 630 faces and 3552 edges and a volume of 49 663 mm³. Its bounding box does not match the nominal design:

Dimension	Nominal design	Audited STEP	Difference
Width (X)	50.00 mm	40.000 mm	−20 %
Length (Y, load axis)	100.00 mm	79.998 mm	−20 %
Depth (Z)	50.00 mm	25.000 mm	−50 %

Table 3. STEP bounding box against the analytical reference design.

Width and length are both 20 % smaller and the depth is halved. The CAD is therefore a scaled, thinner version of the analytical design, and the square side, chamfer and ligament width could not be isolated from the fused solid. Two consequences follow: forces and stresses from the analytical model (which uses the 50 mm depth) cannot be compared directly with the simulation, and any gross stress or modulus must use the solved model's own dimensions. The Poisson's ratio, a ratio of strains, is unaffected.

> **What the audit establishes:** the file is readable, in millimetres, one solid, with the dimensions above.
>
> **What it does not:** the exact a, d, h and end-block length of the CAD — the fused solid has not been decomposed into its features.

2.3 Finite-element simulation

The specimen was solved in ANSYS Mechanical 2025 R1 (Static Structural): lower end fixed, upper end displaced 5.00 mm toward −Y, isotropic linear-elastic TPU (E = 26 MPa, v = 0.45), large deflection on, transverse motion at the loaded end free. Two internal gauges were placed on the undeformed lattice away from the end blocks, each 11.09 mm long: probes 1–2 along the load axis and 3–4 across it.

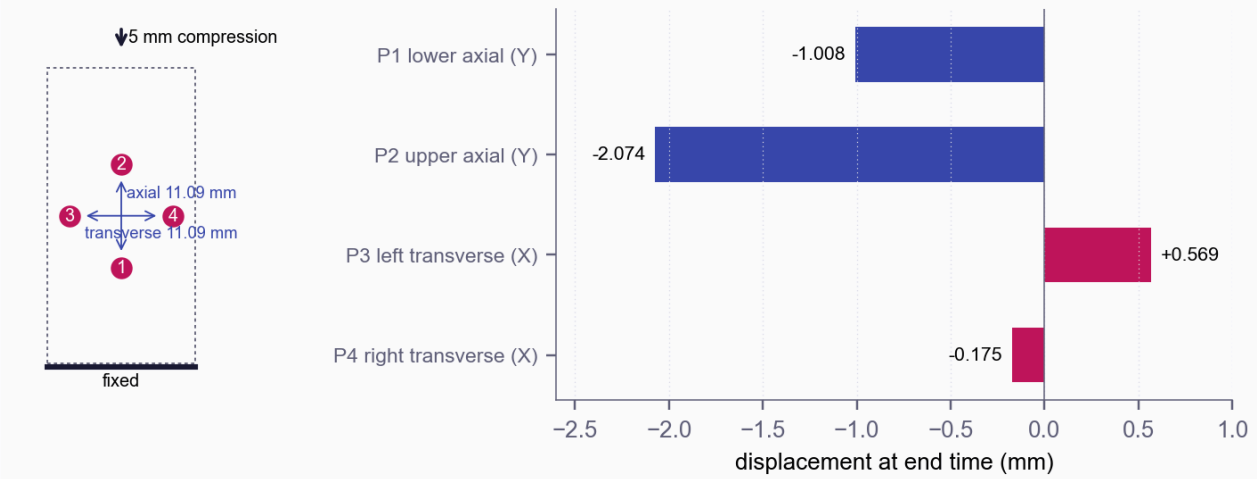


Fig. 16 Gauge layout (left) and probe displacements at the end of the load step (right).

$$\varepsilon_Y = \frac{-2.0737 - (-1.0075)}{11.09} = -9.61\,\%, \qquad \varepsilon_X = \frac{-0.17464 - 0.56920}{11.09} = -6.71\,\%$$

$$\nu_{\text{eff}} = -\frac{\varepsilon_X}{\varepsilon_Y} = -0.698 \tag{7}$$

End-time output	Value
Fixed-support axial reaction	29.895 N
Secant stiffness (reaction / 5 mm)	5.979 N/mm
Maximum total deformation	5.847 mm
X deformation range	−2.137 to +2.074 mm
Maximum von Mises stress	2.974 MPa

Table 4. ANSYS results at 5 mm of compression.

The X-displacement field shows the left side moving right and the right side moving left over most of the lattice — the specimen draws in as it is compressed. Both gauges shorten together: the lattice is auxetic under this load.

2.4 Comparison and discussion

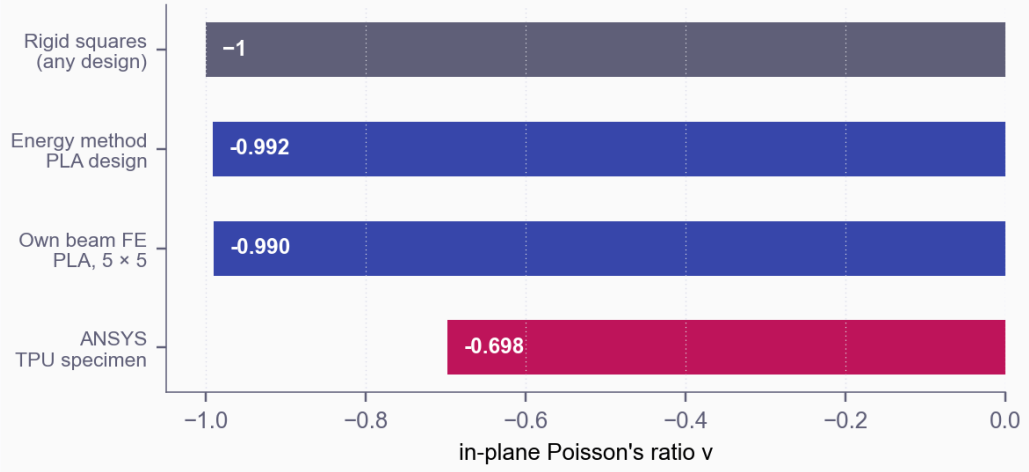


Fig. 17 Poisson's ratio across every description in the strand.

The ideal model predicts v = −1; the compliant-hinge models of the stiff PLA design stay within 1 % of it. The simulated TPU specimen reaches **−0.698**: its transverse strain is about 70 % of its axial strain. Several departures from the ideal mechanism explain the gap, none of which the rigid-square or slender-hinge models include:

- **Stubby ligaments.** With $L_h/h \approx 1.5$, the TPU hinges stretch and shear as well as bend; Eq. (4) shows that any stretching pulls v toward zero.
- **Deformable squares.** In a soft material the squares themselves strain, absorbing some of the motion that would otherwise rotate them.
- **A finite, constrained specimen.** Only six rows tall and three wide, with solid end blocks that hold the boundary rows; the beam FE showed even a stiff 5 × 5 array departing from the periodic result.
- **Large strain.** At 9.6 % gauge strain the response is no longer the small-strain secant value the hinge models give.

The ratio is also a local estimate: it uses face-maximum displacements over two short gauges, and rotation of the probed faces can bias the differences. It describes these gauges at 5 mm of travel, not a material constant.

Stiffness. The simulation gives a secant stiffness of 5.98 N/mm. The hinge model's force estimates (94.7 N for 10 mm of active-region compression) refer to the 50 mm-deep nominal design and a different displacement definition, and the audited CAD is half as deep, so the two are not directly comparable.

2.5 Limitations and next steps

- **Material.** E = 26 MPa and v = 0.45 are provisional TPU values; a linear-elastic law cannot capture hyperelasticity, viscoelasticity or rate dependence.
- **Geometry.** The simulated CAD differs from the analytical design by 20 % in plan and 50 % in depth, and its feature sizes are unverified.
- **Mesh.** One mesh and one load case; peak stress at the ligaments is mesh-sensitive and no convergence study was run.
- **Hinge model.** Not slender ($L_h/h \approx 1.5$); shear, axial stretching and chamfer stress concentrations are omitted.
- **Scope.** Contact at closure, tension, cyclic loading and physical testing are outside what has been shown.

Next steps. Re-run the simulation on CAD matching the analytical design, with a mesh study and gauges over more cell pairs; calibrate a hyperelastic TPU law; extend the load to closure with contact; and add tension, where the hinge model predicts the larger force.

Where the mechanism leads

Rotating units give an isotropic auxetic response from a single repeating motif, which is why they recur in stent-like tubes that expand in every direction at once [6, 9] and in hierarchical metamaterials with programmable stiffness [7]. A soft, ductile rotating-squares lattice that moves elastically through tens of percent of strain — the goal of the TPU redesign — is the enabling step for those uses.

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